



Principles and Anti-Patterns of The Bayes Theorem

Principles

1. Subjective Probabilities:

Estimates from Bayes theory can change depending on hypothesis and other factors; Decision Makers (DMs) need to have an 'internal belief' in what a likely probability should be.

2. Hypothesis/ Prior selection:

Outputs can be strongly affected by the hypotheses chosen and how they are expressed. Among Bayesians, there are two approaches to choosing prior distributions. Some recommend choosing the prior that best represents background information about the problem; in that case the prior is said to be **informative**. The problem with using an informative prior is that people might have different information or interpret it differently. So informative priors might seem arbitrary.

*"The alternative is a so-called **uninformative prior**, which is intended to be as unrestricted as possible, in order to let the data speak for itself. In some cases, you can identify a unique prior that has some desirable property, like representing minimal prior information about the estimated quantity.*

Uninformative priors are appealing because they seem more objective. But I am generally in favor of using informative priors. Why? First, Bayesian analysis is always based on modeling decisions. Choosing the prior is one of those decisions, but it is not the only one, and it might not even be the most subjective. So even if an uninformative prior is more objective, the entire analysis is still subjective."

Downey (2016)



Anti-Patterns

1. Ambiguity:

Not all sources of uncertainty can be quantified – such as the price of copper in 20 years' time OR likelihood of a war – cannot be modelled using standard PDs.

2. The inability to express ignorance is a failing of the system:

There is a requirement that decision makers have an opinion on every question answered. Just providing arbitrary answers may seriously skew the calculated results.

3. The Ellsberg paradox:

Which concerns ambiguity aversion. People prefer known or calculable risks. This depends on the decision maker , and cannot be predicted.